

Complex pre-explosion mass loss in red supergiants, as illuminated by the SN 1998S-like SN 2024cld

Early-time flash ionisation → persistent interaction with disk-like CSM

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FIG 01 • Deep NOT/ALFOSC gri image of SN 2024cld



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KEY RESULT

SN 2024cld was discovered just **11h post-explosion** via the GOTO-FAST program, likely the earliest of the elusive 98S-like SNe. Continuous monitoring over the next 200d revealed complex, aspherical CSM driving enduring interaction, attributable to a **disk of material shed by binary interaction**, with spectroscopy and polarimetry pinning the geometry. A prodigious dust component at late time further underscores **disk-interacting SNe as potent dust factories** – with dust properties and formation to be probed in SN 2024cld via **awarded JWST Cycle 5 time**.

Continuous coverage: from 11h post=explosion to 200d (and beyond)

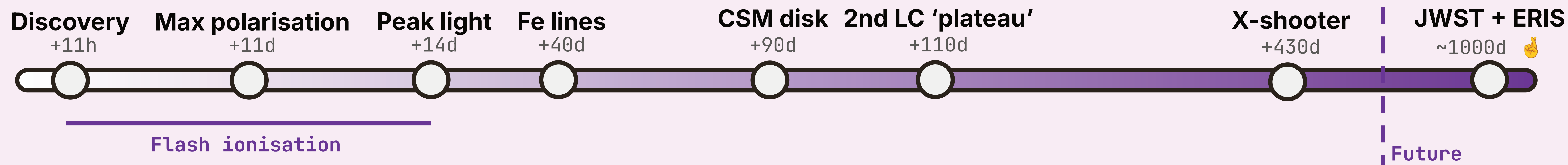


FIG 02 • Timeline

01 Discovery and extended flash-ionisation phase



Discovered by GOTO and classified on the INT just **11 hours post-explosion**, as part of the **GOTO-FAST** transient follow-up program – pairing a tailored high-cadence sky survey, with visitor mode allocations for responsive classifications.

SN 2024cld initially looked like an ordinary flash-ionised (FI) SN II: but with a long-duration FI phase, extended rise to peak, and continued narrow emission even after two weeks, interaction likely dominates the evolution.

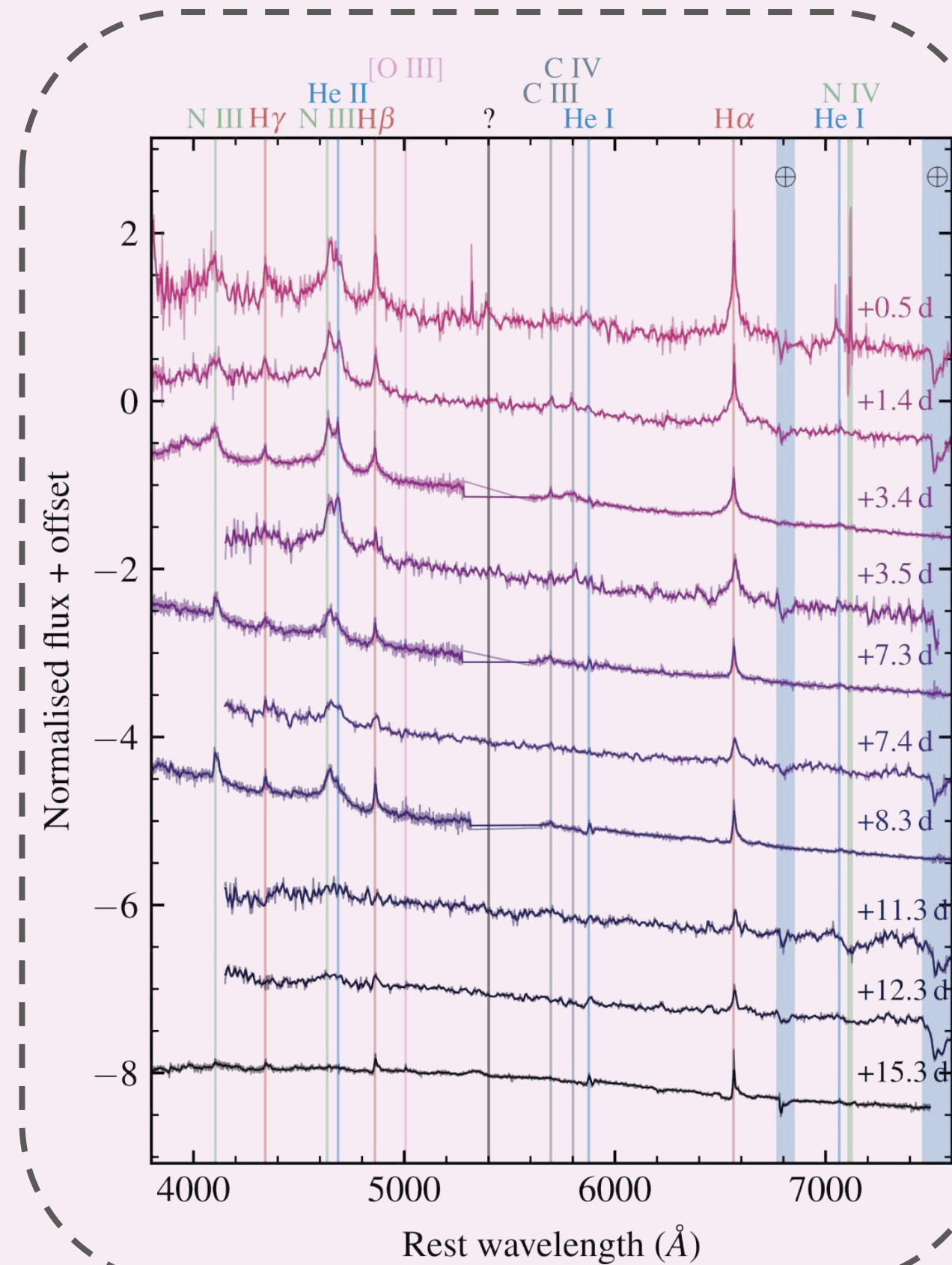


FIG 03 • Early-time flash features of SN 2024cld, with He II, C III/IV, and NIII

02 An emerging multi-component CSM

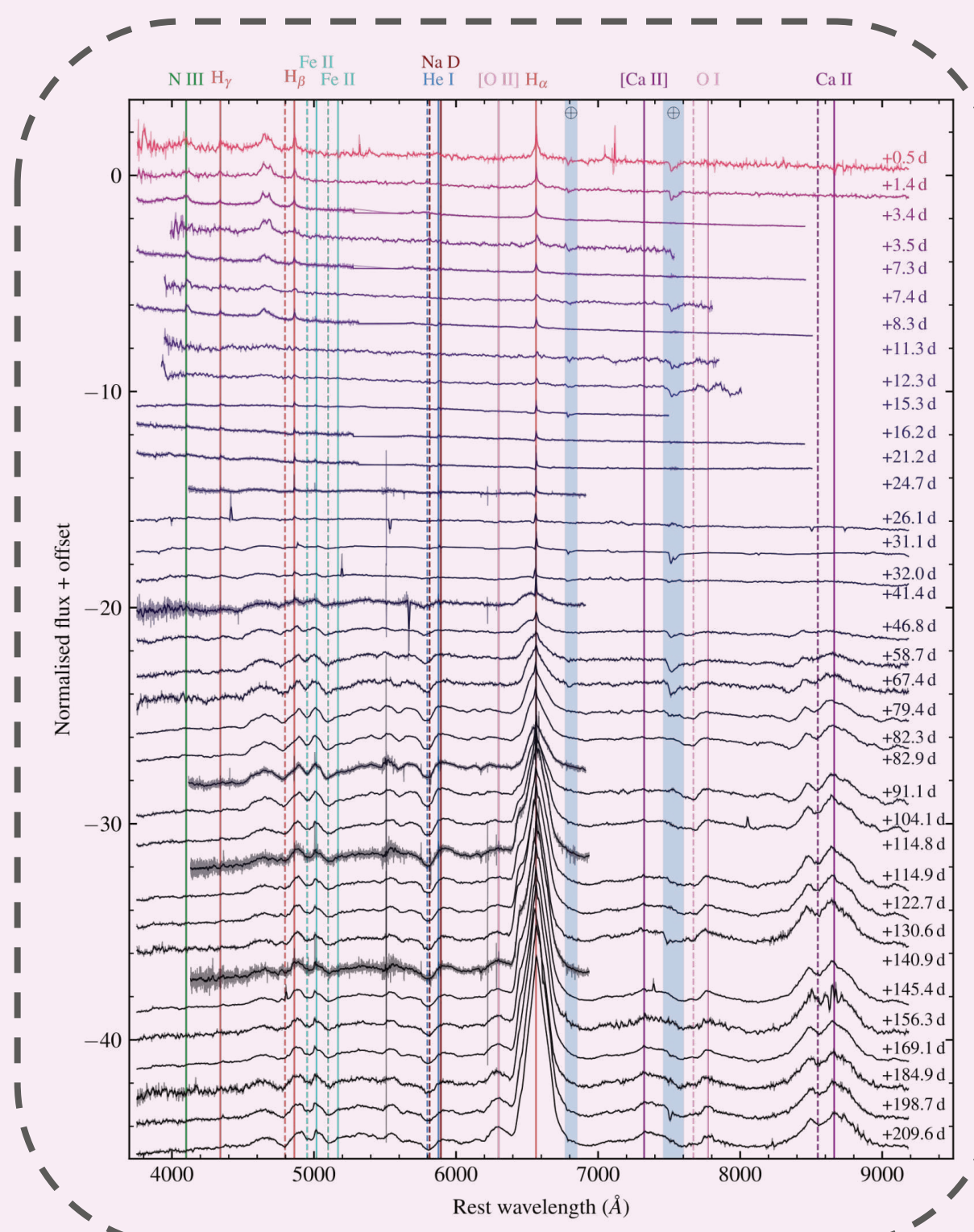


FIG 04 • Full spectral series of SN 2024cld, from 11h post-explosion to +200d

FIG 05 • High-resolution spectroscopy reveals a narrow P Cygni profile, associated with pre-shock wind

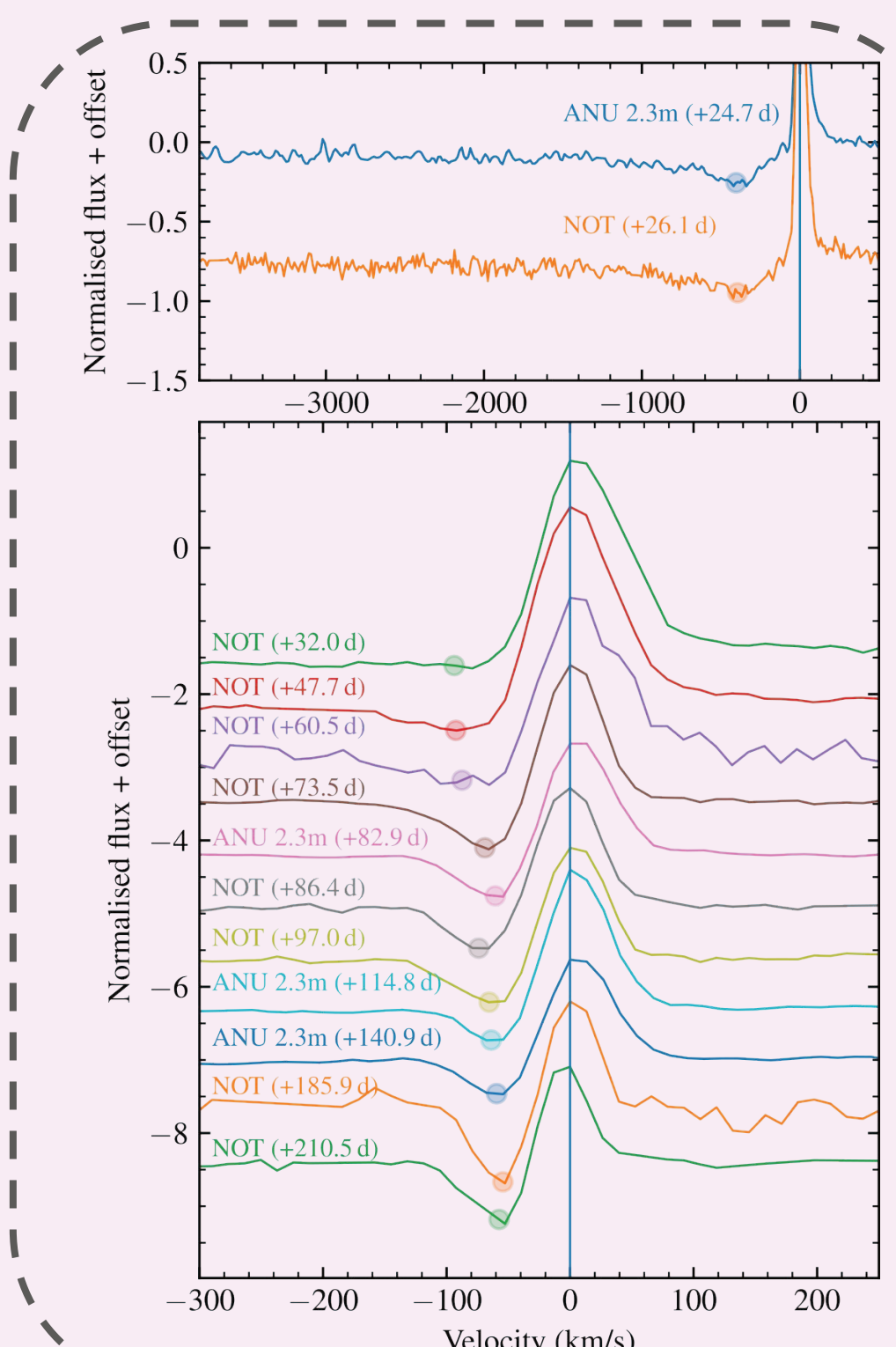
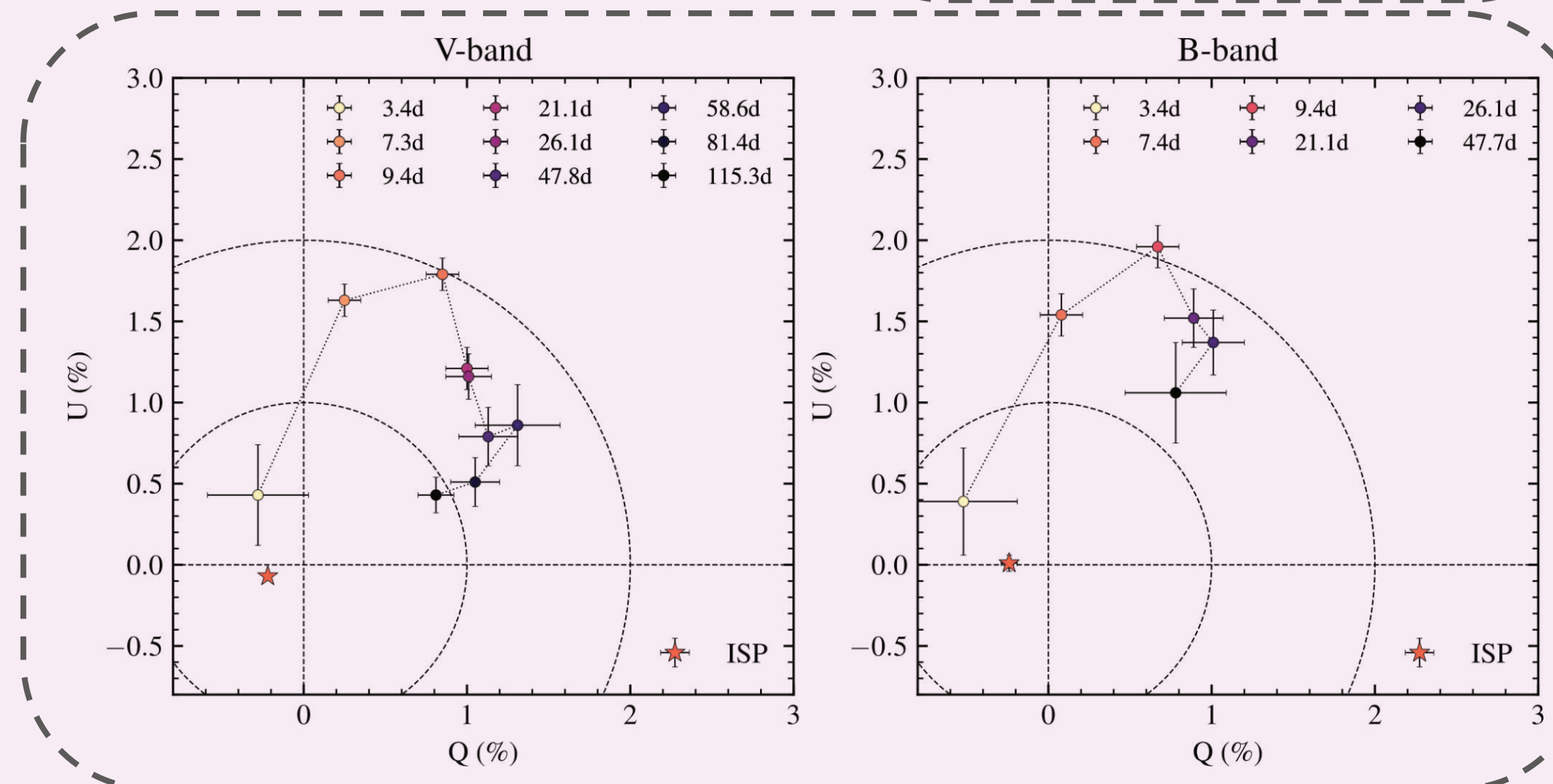


FIG 06 • Polarisation evolution of SN 2024cld, showing a rise to a peak of 2% around 10 days post-explosion, followed by a slow decline and rotation.



03 Light curve evolution

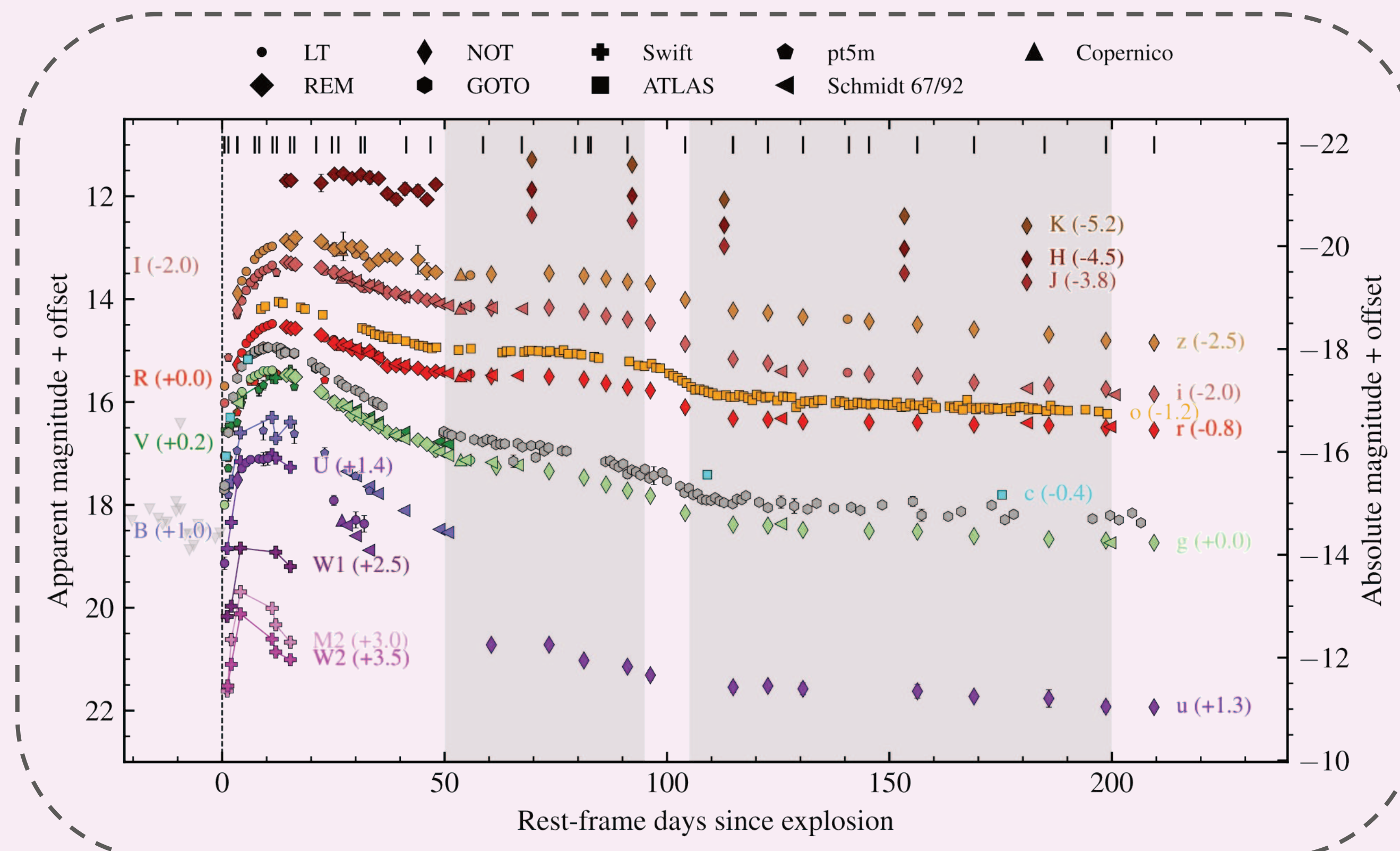


FIG 07 • Light curve for SN 2024cld. The SN reaches a peak absolute magnitude of $g=-17.58$ at **14.6 days** post-explosion. The late-time evolution is entirely driven by CSM interaction

05 Scenario

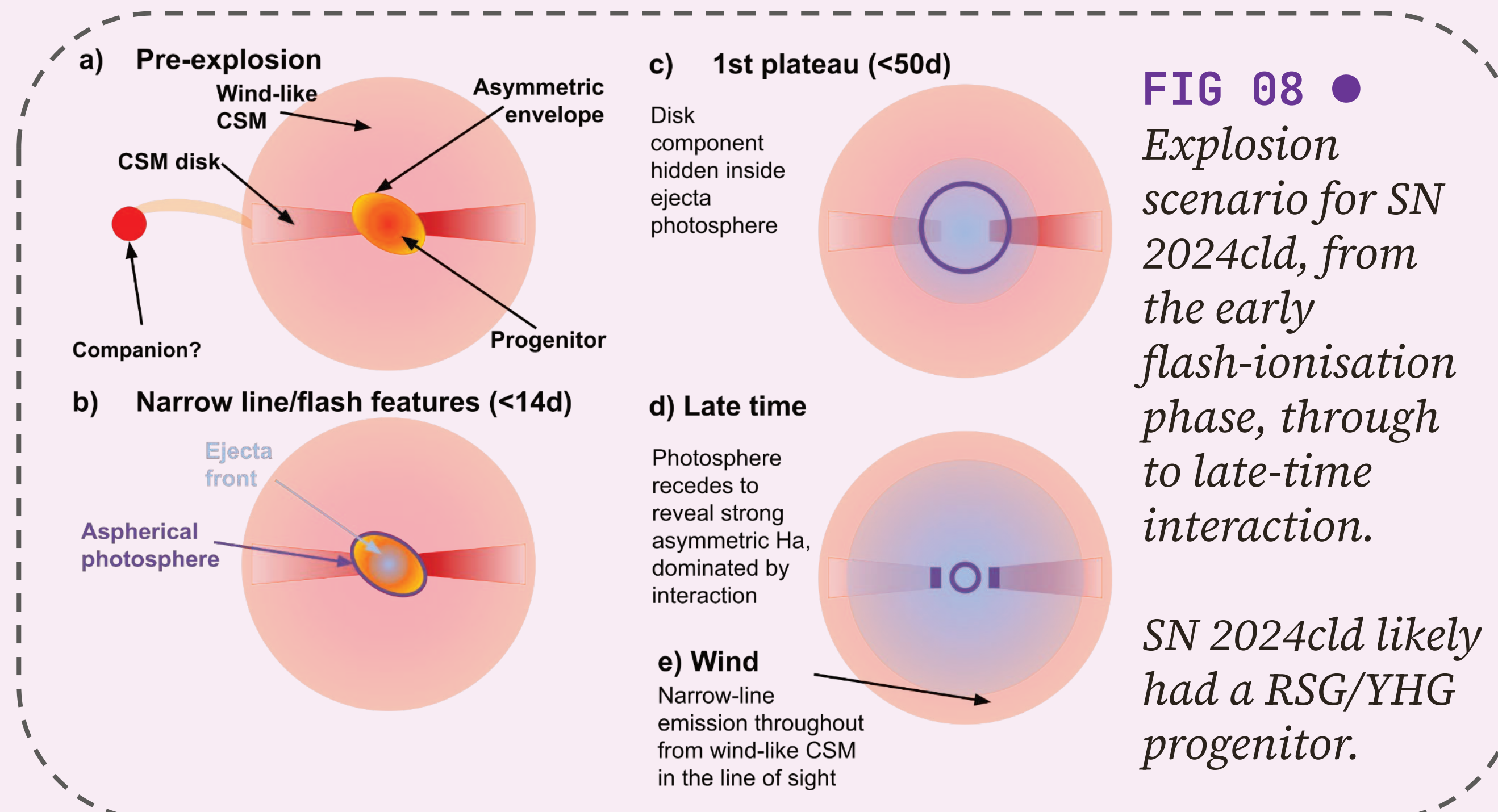


FIG 08 • Explosion scenario for SN 2024cld, from the early flash-ionisation phase, through to late-time interaction. SN 2024cld likely had a RSG/YHG progenitor.

06 Continued evolution

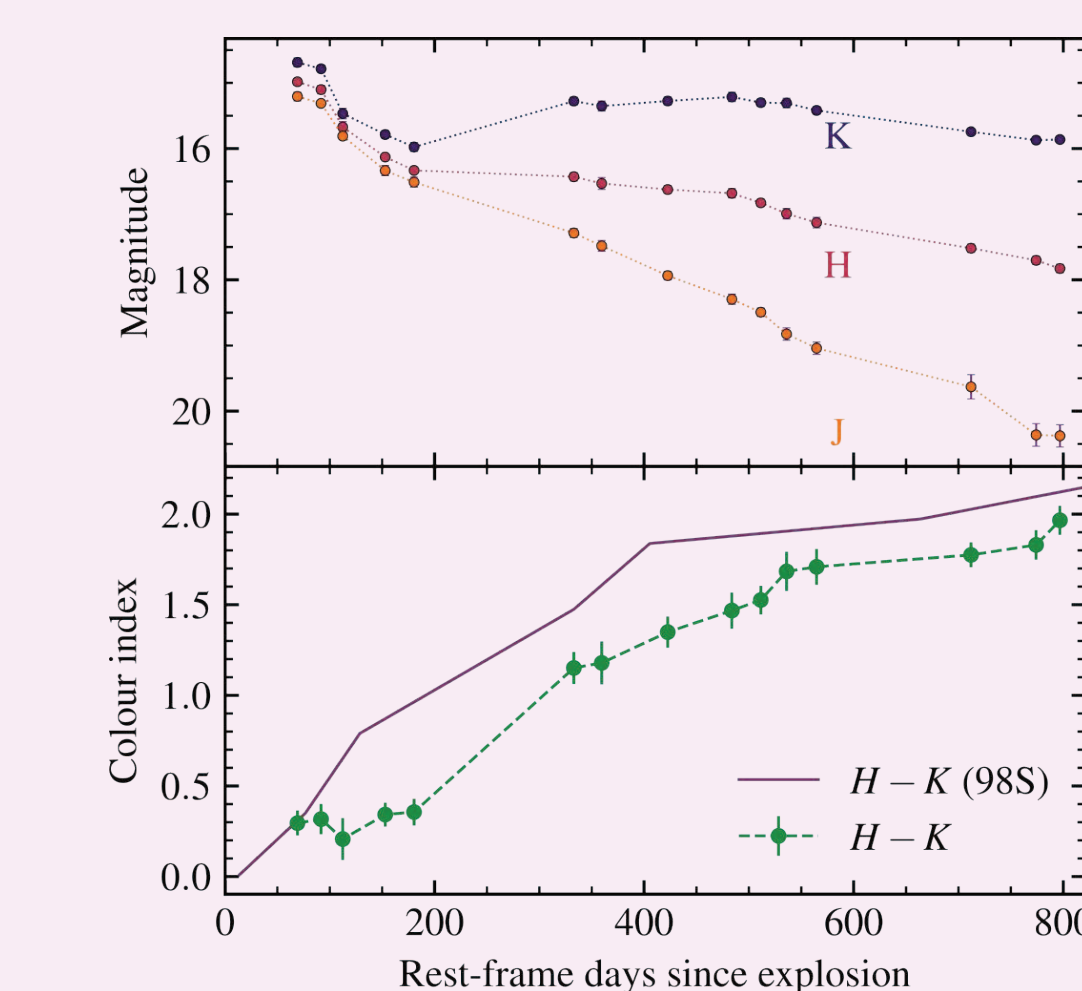


FIG 09 • From +400d onwards, SN 2024cld has shown a clear K-band excess, and dust-related features in a late time NIR spectrum. Awarded **X-shooter**, **ERIS**, and **JWST time** will constrain the dust properties and assess the dust yield of this exotic family of SNe.

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